Kulachart Parnduangkeaw

Machine Learning · Full-Stack Web & Mobile · Computer Engineering, SWU

Pathum Thani, Thailand · +66 80 091 2939 · kulachart@kulac.dev · kulac.dev · linkedin.com/in/kulachart-parnduangkeaw-247b47395 · github.com/KhaNomThai



Key results **Data Science Intern** Botnoi Group, 2026 / **Student Leader** Dept. of Computer Engineering, SWU, 2024 – Present / **Top 30** KBTG Hackathon 2026 / **Top 24** NTT DATA Innovation Challenge 2026 / **~300 users** SWU Run web app, 2026

Summary

Computer Engineering student at SWU (expected 2028) who builds AI systems end to end, from the model to the app people use. Data Science Intern at Botnoi Group (2026), building demand forecasting and MLOps for SME clients. Shipped a skin-cancer classifier that runs in the browser, a Bedrock agent on serverless AWS, and a React app about 300 runners used; top 30 at KBTG Hackathon and top 24 at NTT DATA Challenge 2026. Department student leader who has taught robotics to high-school students. Seeking an internship in AI / ML, Data Science or Full-Stack Development.

Experience

Data Science Intern — Botnoi Group · May – Jul 2026

- Built product demand forecasting for SME clients: data cleaning, feature engineering, model training and evaluation.
- Set up the MLOps pipeline to train and deploy the models, and built dashboards for the client companies.
- Engineered features that improved model accuracy on real-world business data.
- Worked with engineering and product teams to deliver AI-powered solutions and data-driven insights.

Projects

AI **KaTang Cash-flow Co-pilot** — KBTG Hackathon 2026 · Team of 3 · Data science · 2026

- Built the gradient-boosting spending forecast and Monte Carlo prototype (simulated data) behind a safe daily spending limit for variable-income first jobbers; **pitched in the top-30 round**.

Python · scikit-learn · Pandas · Machine Learning · Next.js · TypeScript

AI · Web **WELLYNC** — NTT DATA Digital Innovation Challenge 2026 · Team of 3 · Web & LLM dev · 2026

- Built a Bedrock agent on serverless AWS (Lambda, DynamoDB, Cognito) that picks role-scoped tools to draft care plans for doctor review in a chronic-care prototype; **reached the top 24 teams**.
- Built the Next.js web app where patients, caregivers and doctors share one consent-checked record.

Next.js · TypeScript · Tailwind CSS · AWS · Amazon Bedrock · Generative AI

AI · Web **Derma AI: Skin Cancer Screening** — CPE310 Healthcare AI · Solo project · 2026

- Fine-tuned ImageNet-pretrained ResNet50, EfficientNet-B3 and DenseNet121 on ISIC 2019 (25,331 images, 8 classes) with class-weighted loss for the imbalanced classes.
- Split by lesion (StratifiedGroupKFold) so no test lesion is seen in training; tuned each model's cancer threshold on validation to target 90% sensitivity.
- Best model: **0.89 AUC-ROC** and 88.7% top-2 accuracy on 3,619 unseen test images; exported to ONNX so it runs in the browser and images never leave the device.

Python · PyTorch · scikit-learn · Pandas · Deep Learning · Computer Vision · ONNX Runtime · React

AI · Web **Smart Flood AI** — DOH Hackathon 2026 · Team of 3 · Web & AI dev · 2026

- Built the zero-shot Grounding DINO pipeline that finds cars and wheels in highway CCTV, and the logic to **estimate water depth** from how far the wheels are submerged, with no labeled training data.
- Built the Next.js command-center dashboard on the team's FastAPI backend, with flood-aware routing and LINE OA alerts.

Python · PyTorch · OpenCV · Grounding DINO · FastAPI · Next.js · Firebase · Computer Vision

Web **SWU Run Together 2026** — SWU Athletics Club · Team of 2 · Developer · 2026

- Built the React web app and its live Google Sheets connection for a university running event: runners search by phone number to get a digital BIB with package, shirt size and payment status; **used by about 300 runners**.

React · JavaScript · Vite · Google Sheets

Mobile **BrainFit** — Client project · Team of 5 · 2025

- Cross-platform React Native (Expo) brain-training app for iOS and Android with memory and logic mini-games, commissioned by nursing students to support people with conditions such as Alzheimer's; **delivered to the client for their research**.
- Built the sign-in flow connected to the Google Sheets backend (via Apps Script) and parts of the app screens.

React Native · Expo · JavaScript · Google Sheets

Data **Peak Electricity Day Alert** — CPE312 Data Science capstone · 2026

- Forecast a week ahead which days hit peak demand from 2015–2020 Panama load and weather data, with leakage-safe daily features and a time-based train/test split.
- XGBoost reached **84% recall on peak days vs 73%** for a same-day-last-week baseline, at higher precision (78% vs 73%); also compared Logistic Regression.

Python · XGBoost · scikit-learn · Pandas · Matplotlib · Machine Learning

Data **Employment Data Gap Analysis** — Research project · 2025

- Random Forest on 100,000 placement records vs a survey of 80 CPE students: students weighted internships and portfolio above the model (**31% vs 24%**) and technical skills below it (33% vs 39%).

Python · scikit-learn · Pandas · Matplotlib · Seaborn · Machine Learning · Data Analysis

Web **Prim's Algorithm Visualizer** — CPE101 course project · 2024

- Interactive D3.js tool that builds a Minimum Spanning Tree step by step while the total cost updates in real time.

JavaScript · D3.js · HTML · CSS

Web **This Portfolio** — Web App · 2026

- One typed data file drives this website and its PDF CV; React 19 and Tailwind CSS v4, with lint, data and CV-sync checks in CI on every push.

React · TypeScript · Tailwind CSS · Framer Motion

Leadership & Teaching

Computer Engineering Student Leader — Dept. of Computer Engineering, SWU · 2024 – Present

- Represent Computer Engineering students at faculty meetings and university events as the main link between students and faculty.
- Plan and run workshops, meetings, and student-development programs for the department.
- Taught micro:bit programming to high-school students at a faculty activity and emceed a university event (2025).

Robotics Teaching Assistant — Srinakharinwirot University · 2026

- Taught robotics and programming fundamentals to high-school students in hands-on sessions.
- Prepared and maintained the robotics equipment and ran live robot-programming demos.
- Guided students through club activities to build their problem-solving skills.

Metaverse Presentation Assistant — Srinakharinwirot University · 2025

- Assisted with a lecture on Metaverse technology, explaining how VR, AR, and AI work together in virtual worlds.
- Built interactive slides and live demos to keep the audience engaged.

Research Presentation Assistant — Dept. of Computer Engineering, SWU · 2025

- Helped a faculty member present a research poster at Chulachomklao Royal Military Academy on classifying Acanthamoeba cysts in microscope images with pretrained CNN features and an SVM classifier.

Education

Bachelor of Engineering in Computer Engineering — Srinakharinwirot University (SWU) · 2024 – Present · Expected 2028
Relevant coursework: Software Development, Database Systems, Artificial Intelligence, Healthcare AI Systems, Computer Architecture, Electronics

Honors & Certifications

- Top 30, KBTG Hackathon 2026 (KaTang Cash-flow Co-pilot)
- Top 24, NTT DATA Digital Innovation Challenge 2026 (WELLYNC)
- Bangkok Insurance Scholarship — Bangkok Insurance, 2025
- Google Cloud, Big Data & GenAI Training — Google Cloud, 2026

Skills

AI / ML: PyTorch, scikit-learn, XGBoost, OpenCV, Grounding DINO, ONNX Runtime, Amazon Bedrock (LLM agents)
Methods: Demand forecasting, transfer learning, time-based validation, threshold tuning, Monte Carlo simulation, MLOps
Data: Python, Pandas, Matplotlib, Seaborn, Google Colab
Web: React, Next.js, TypeScript, JavaScript, Tailwind CSS, D3.js, HTML/CSS, Vite
Mobile: React Native, Expo
Backend & cloud: FastAPI, AWS (Lambda, API Gateway, DynamoDB, Cognito), Firebase, Google Apps Script, Google Cloud
Tools: Git, GitHub Actions, Vercel
Languages: Thai (native), English (intermediate)